PRAKTIKUM BASIS DATA

MODUL 2

(MODUL 2)

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

JURNAL1

1. Buatlah setiap tabel diatas, tuliskan script SQL nya dan sertakan tangkapan layarnya.

   Screenshot :

```
SQL> --nim: 103122430043
SQL> CREATE TABLE Film (
  2      id_film NUMBER PRIMARY KEY,
  3      judul VARCHAR2(255),
  4      sinopsis CLOB,
  5      tahun NUMBER(4),
  6      durasi NUMBER
  7  );

Table created.

SQL> CREATE TABLE Theater (
  2      id_theater NUMBER PRIMARY KEY,
  3      harga NUMBER(10, 2),
  4      kapasitas NUMBER,
  5      kelas VARCHAR2(50)
  6  );

Table created.

SQL> CREATE TABLE Inventaris (
  2      id_inventaris NUMBER PRIMARY KEY,
  3      id_theater NUMBER,
  4      nomor_kursi VARCHAR2(10),
  5      CONSTRAINT fk_theater_inv FOREIGN KEY (id_theater)
  6          REFERENCES Theater(id_theater)
  7  );

Table created.

SQL> CREATE TABLE Member (
  2      id_member NUMBER PRIMARY KEY,
  3      nama VARCHAR2(100),
  4      no_hp VARCHAR2(20),
  5      tgl_lahir DATE,
  6      email VARCHAR2(100)
  7  );

Table created.
```

```
SQL> CREATE TABLE Jadwal (
  2      id_jadwal NUMBER PRIMARY KEY,
  3      id_film NUMBER,
  4      id_theater NUMBER,
  5      periode_start DATE,
  6      periode_end DATE,
  7      CONSTRAINT fk_film_jadwal FOREIGN KEY (id_film)
  8          REFERENCES Film(id_film),
  9      CONSTRAINT fk_theater_jadwal FOREIGN KEY (id_theater)
 10          REFERENCES Theater(id_theater)
 11  );

Table created.

SQL> CREATE TABLE Transaksi (
  2      kode_pemesanan VARCHAR2(20) PRIMARY KEY,
  3      id_jadwal NUMBER,
  4      id_member NUMBER,
  5      status VARCHAR2(50),
  6      tanggal DATE,
  7      total_harga NUMBER(10, 2),
  8      CONSTRAINT fk_jadwal_trx FOREIGN KEY (id_jadwal)
  9          REFERENCES Jadwal(id_jadwal),
 10      CONSTRAINT fk_member_trx FOREIGN KEY (id_member)
 11          REFERENCES Member(id_member)
 12  );

Table created.
```

```
SQL> desc transaksi
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 KODE_PEMESANAN                            NOT NULL VARCHAR2(20)
 ID_JADWAL                                          NUMBER
 ID_MEMBER                                          NUMBER
 STATUS                                             VARCHAR2(50)
 TANGGAL                                            DATE
 TOTAL_HARGA                                        NUMBER(10,2)

SQL> desc jadwal
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ID_JADWAL                                 NOT NULL NUMBER
 ID_FILM                                            NUMBER
 ID_THEATER                                         NUMBER
 PERIODE_START                                      DATE
 PERIODE_END                                        DATE

SQL> desc inventaris
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ID_INVENTARIS                             NOT NULL NUMBER
 ID_THEATER                                         NUMBER
 NOMOR_KURSI                                        VARCHAR2(10)
```

```
SQL> desch theater
SP2-0734: unknown command beginning "desch thea..." - rest of line ignored.
SQL> desc theater
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ID_THEATER                                NOT NULL NUMBER
 HARGA                                              NUMBER(10,2)
 KAPASITAS                                          NUMBER
 KELAS                                              VARCHAR2(50)

SQL> desc film
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ID_FILM                                   NOT NULL NUMBER
 JUDUL                                              VARCHAR2(255)
 SINOPSIS                                           CLOB
 TAHUN                                              NUMBER(4)
 DURASI                                             NUMBER
```

Jawaban :

```
SQL> CREATE TABLE Film (
  2     id_film NUMBER PRIMARY KEY,
  3     judul VARCHAR2(255),
  4     sinopsis CLOB,
  5     tahun NUMBER(4),
  6     durasi NUMBER
  7  );
SQL> CREATE TABLE Theater (
  2     id_theater NUMBER PRIMARY KEY,
  3     harga NUMBER(10, 2),
  4     kapasitas NUMBER,
  5     kelas VARCHAR2(50)
  6  );
SQL> CREATE TABLE Inventaris (
  2     id_inventaris NUMBER PRIMARY KEY,
  3     id_theater NUMBER,
  4     nomor_kursi VARCHAR2(10),
  5     CONSTRAINT fk_theater_inv FOREIGN KEY (id_theater)
  6         REFERENCES Theater(id_theater)
  7  );
SQL> CREATE TABLE Member (
  2     id_member NUMBER PRIMARY KEY,
  3     nama VARCHAR2(100),
  4     no_hp VARCHAR2(20),
  5     tgl_lahir DATE,
  6     email VARCHAR2(100)
  7  );
SQL> CREATE TABLE Jadwal (
  2     id_jadwal NUMBER PRIMARY KEY,
  3     id_film NUMBER,
```

```
  4     id_theater NUMBER,
  5     periode_start DATE,
  6     periode_end DATE,
  7     CONSTRAINT fk_film_jadwal FOREIGN KEY (id_film)
  8        REFERENCES Film(id_film),
  9     CONSTRAINT fk_theater_jadwal FOREIGN KEY (id_theater)
 10        REFERENCES Theater(id_theater)
 11  );
SQL> CREATE TABLE Transaksi (
  2     kode_pemesanan VARCHAR2(20) PRIMARY KEY,
  3     id_jadwal NUMBER,
  4     id_member NUMBER,
  5     status VARCHAR2(50),
  6     tanggal DATE,
  7     total_harga NUMBER(10, 2),
  8     CONSTRAINT fk_jadwal_trx FOREIGN KEY (id_jadwal)
  9        REFERENCES Jadwal(id_jadwal),
 10     CONSTRAINT fk_member_trx FOREIGN KEY (id_member)
 11        REFERENCES Member(id_member)
 12  );
```

Penjelasan :
Saya menggunakan CREATE TABLE dengan PRIMARY KEY untuk identitas unik dan FOREIGN KEY untuk menghubungkan tabel (misalnya menghubungkan Film ke Jadwal). Urutan pembuatan sangat penting, buat tabel "Induk" (Film, Theater, Member) baru kemudian tabel "Anak" (Jadwal, Transaksi).

2. Lakukan insert data pada ke-enam tabel, sertakan tangkapan layar dan keterangannya.

   Screenshot :

```
SQL> INSERT ALL
  2    INTO Film (id_film, judul, sinopsis, tahun, durasi) VALUES (1, 'Your Name', 'Remaja bertukar tubuh', 2016, 106)
  3    INTO Film (id_film, judul, sinopsis, tahun, durasi) VALUES (2, 'Weathering With You', 'Gadis pengendali cuaca', 2019, 112)
  4    INTO Film (id_film, judul, sinopsis, tahun, durasi) VALUES (3, 'Suzume', 'Pintu bencana', 2022, 122)
  5    INTO Film (id_film, judul, sinopsis, tahun, durasi) VALUES (4, '5 Centimeters per Second', 'Cinta dan jarak', 2007, 63)
  6    INTO Film (id_film, judul, sinopsis, tahun, durasi) VALUES (5, 'The Garden of Words', 'Pertemuan saat hujan', 2013, 46)
  7  SELECT * FROM dual;

5 rows created.

SQL> INSERT ALL
  2    INTO Theater (id_theater, harga, kapasitas, kelas) VALUES (101, 45000, 50, 'Reguler')
  3    INTO Theater (id_theater, harga, kapasitas, kelas) VALUES (102, 50000, 40, 'Reguler Plus')
  4    INTO Theater (id_theater, harga, kapasitas, kelas) VALUES (103, 75000, 30, 'Dolby')
  5    INTO Theater (id_theater, harga, kapasitas, kelas) VALUES (104, 100000, 20, 'Premiere')
  6    INTO Theater (id_theater, harga, kapasitas, kelas) VALUES (105, 150000, 15, 'Velvet')
  7  SELECT * FROM dual;

5 rows created.

SQL> INSERT ALL
  2    INTO Inventaris (id_inventaris, id_theater, nomor_kursi) VALUES (1, 101, 'A1')
  3    INTO Inventaris (id_inventaris, id_theater, nomor_kursi) VALUES (2, 101, 'A2')
  4    INTO Inventaris (id_inventaris, id_theater, nomor_kursi) VALUES (3, 102, 'B1')
  5    INTO Inventaris (id_inventaris, id_theater, nomor_kursi) VALUES (4, 103, 'C1')
  6    INTO Inventaris (id_inventaris, id_theater, nomor_kursi) VALUES (5, 104, 'D1')
  7  SELECT * FROM dual;

5 rows created.

SQL> INSERT ALL
  2    INTO Member (id_member, nama, no_hp, tgl_lahir, email) VALUES (1, 'Taki', '0811', TO_DATE('1999-12-01','YYYY-MM-DD'), 'taki@mail.com')
  3    INTO Member (id_member, nama, no_hp, tgl_lahir, email) VALUES (2, 'Mitsuha', '0812', TO_DATE('1997-04-20','YYYY-MM-DD'), 'mitsuha@mail.com')
  4    INTO Member (id_member, nama, no_hp, tgl_lahir, email) VALUES (3, 'Hodaka', '0813', TO_DATE('2005-01-10','YYYY-MM-DD'), 'hodaka@mail.com')
  5    INTO Member (id_member, nama, no_hp, tgl_lahir, email) VALUES (4, 'Suzume', '0814', TO_DATE('2006-05-24','YYYY-MM-DD'), 'suzume@mail.com')
  6    INTO Member (id_member, nama, no_hp, tgl_lahir, email) VALUES (5, 'Takaki', '0815', TO_DATE('1985-03-15','YYYY-MM-DD'), 'takaki@mail.com')
  7  SELECT * FROM dual;

5 rows created.
```

```
SQL> INSERT ALL
  2    INTO Jadwal (id_jadwal, id_film, id_theater, periode_start, periode_end) VALUES (501, 1, 101, TO_DATE('2026-03-10 13:00','YYYY-MM-DD HH24:MI'), TO_DA
TE('2026-03-10 14:46','YYYY-MM-DD HH24:MI'))
  3    INTO Jadwal (id_jadwal, id_film, id_theater, periode_start, periode_end) VALUES (502, 2, 102, TO_DATE('2026-03-10 15:00','YYYY-MM-DD HH24:MI'), TO_DA
TE('2026-03-10 16:52','YYYY-MM-DD HH24:MI'))
  4    INTO Jadwal (id_jadwal, id_film, id_theater, periode_start, periode_end) VALUES (503, 3, 103, TO_DATE('2026-03-10 17:00','YYYY-MM-DD HH24:MI'), TO_DA
TE('2026-03-10 19:02','YYYY-MM-DD HH24:MI'))
  5    INTO Jadwal (id_jadwal, id_film, id_theater, periode_start, periode_end) VALUES (504, 4, 104, TO_DATE('2026-03-11 19:00','YYYY-MM-DD HH24:MI'), TO_DA
TE('2026-03-11 20:03','YYYY-MM-DD HH24:MI'))
  6    INTO Jadwal (id_jadwal, id_film, id_theater, periode_start, periode_end) VALUES (505, 5, 105, TO_DATE('2026-03-11 21:00','YYYY-MM-DD HH24:MI'), TO_DA
TE('2026-03-11 21:46','YYYY-MM-DD HH24:MI'))
  7  SELECT * FROM dual;

5 rows created.

SQL> INSERT ALL
  2    INTO Transaksi (kode_pemesanan, id_jadwal, id_member, status, tanggal, total_harga) VALUES ('TX01', 501, 1, 'LUNAS', TO_DATE('2026-03-08','YYYY-MM-DD
'), 45000)
  3    INTO Transaksi (kode_pemesanan, id_jadwal, id_member, status, tanggal, total_harga) VALUES ('TX02', 502, 2, 'LUNAS', TO_DATE('2026-03-08','YYYY-MM-DD
'), 50000)
  4    INTO Transaksi (kode_pemesanan, id_jadwal, id_member, status, tanggal, total_harga) VALUES ('TX03', 503, 3, 'LUNAS', TO_DATE('2026-03-08','YYYY-MM-DD
'), 75000)
  5    INTO Transaksi (kode_pemesanan, id_jadwal, id_member, status, tanggal, total_harga) VALUES ('TX04', 504, 4, 'PROSES', TO_DATE('2026-03-08','YYYY-MM-D
D'), 100000)
  6    INTO Transaksi (kode_pemesanan, id_jadwal, id_member, status, tanggal, total_harga) VALUES ('TX05', 505, 5, 'LUNAS', TO_DATE('2026-03-08','YYYY-MM-DD
'), 150000)
  7  SELECT * FROM dual;

5 rows created.
```

```
SQL> select * from film
  2  ;

   ID_FILM JUDUL                    SINOPSIS                           TAHUN     DURASI
---------- ------------------------ ------------------------------ ---------- ----------
         1 Your Name                Remaja bertukar tubuh               2016        106
         2 Weathering With You      Gadis pengendali cuaca              2019        112
         3 Suzume                   Pintu bencana                       2022        122
         4 5 Centimeters per Second Cinta dan jarak                     2007         63
         5 The Garden of Words      Pertemuan saat hujan                2013         46
SQL> select * from transaksi;

KODE_PEMES  ID_JADWAL  ID_MEMBER STATUS     TANGGAL   TOTAL_HARGA
---------- ---------- ---------- ---------- --------- -----------
TX01              501          1 LUNAS      08-MAR-26       45000
TX02              502          2 LUNAS      08-MAR-26       50000
TX03              503          3 LUNAS      08-MAR-26       75000
TX04              504          4 PROSES     08-MAR-26      100000
TX05              505          5 LUNAS      08-MAR-26      150000
```

Jawaban :

SQL> INSERT ALL

2 INTO Film (id_film, judul, sinopsis, tahun, durasi) VALUES (1, 'Your Name', 'Remaja bertukar tubuh', 2016, 106)

3 INTO Film (id_film, judul, sinopsis, tahun, durasi) VALUES (2, 'Weathering With You', 'Gadis pengendali cuaca', 2019, 112)

4 INTO Film (id_film, judul, sinopsis, tahun, durasi) VALUES (3, 'Suzume', 'Pintu bencana', 2022, 122)

```
  5    INTO Film (id_film, judul, sinopsis, tahun, durasi) VALUES (4, '5 Centimeters
per Second', 'Cinta dan jarak', 2007, 63)
  6    INTO Film (id_film, judul, sinopsis, tahun, durasi) VALUES (5, 'The Garden
of Words', 'Pertemuan saat hujan', 2013, 46)
  7  SELECT * FROM dual;
SQL> INSERT ALL
  2    INTO Theater (id_theater, harga, kapasitas, kelas) VALUES (101, 45000, 50,
'Reguler')
  3    INTO Theater (id_theater, harga, kapasitas, kelas) VALUES (102, 50000, 40,
'Reguler Plus')
  4    INTO Theater (id_theater, harga, kapasitas, kelas) VALUES (103, 75000, 30,
'Dolby')
  5    INTO Theater (id_theater, harga, kapasitas, kelas) VALUES (104, 100000, 20,
'Premiere')
  6    INTO Theater (id_theater, harga, kapasitas, kelas) VALUES (105, 150000, 15,
'Velvet')
  7  SELECT * FROM dual;
SQL> INSERT ALL
  2    INTO Inventaris (id_inventaris, id_theater, nomor_kursi) VALUES (1, 101,
'A1')
  3    INTO Inventaris (id_inventaris, id_theater, nomor_kursi) VALUES (2, 101,
'A2')
  4    INTO Inventaris (id_inventaris, id_theater, nomor_kursi) VALUES (3, 102,
'B1')
  5    INTO Inventaris (id_inventaris, id_theater, nomor_kursi) VALUES (4, 103,
'C1')
  6    INTO Inventaris (id_inventaris, id_theater, nomor_kursi) VALUES (5, 104,
'D1')
  7  SELECT * FROM dual;
SQL> INSERT ALL
  2    INTO Member (id_member, nama, no_hp, tgl_lahir, email) VALUES (1, 'Taki',
'0811', TO_DATE('1999-12-01','YYYY-MM-DD'), 'taki@mail.com')
  3    INTO Member (id_member, nama, no_hp, tgl_lahir, email) VALUES (2,
'Mitsuha', '0812', TO_DATE('1997-04-20','YYYY-MM-DD'), 'mitsuha@mail.com')
  4    INTO Member (id_member, nama, no_hp, tgl_lahir, email) VALUES (3,
'Hodaka', '0813', TO_DATE('2005-01-10','YYYY-MM-DD'), 'hodaka@mail.com')
  5    INTO Member (id_member, nama, no_hp, tgl_lahir, email) VALUES (4,
'Suzume', '0814', TO_DATE('2006-05-24','YYYY-MM-DD'), 'suzume@mail.com')
```

```
  6    INTO Member (id_member, nama, no_hp, tgl_lahir, email) VALUES (5,
'Takaki', '0815', TO_DATE('1985-03-15','YYYY-MM-DD'), 'takaki@mail.com')
  7  SELECT * FROM dual;
SQL> INSERT ALL
  2    INTO Jadwal (id_jadwal, id_film, id_theater, periode_start, periode_end)
VALUES (501, 1, 101, TO_DATE('2026-03-10 13:00','YYYY-MM-DD HH24:MI'),
TO_DATE('2026-03-10 14:46','YYYY-MM-DD HH24:MI'))
  3    INTO Jadwal (id_jadwal, id_film, id_theater, periode_start, periode_end)
VALUES (502, 2, 102, TO_DATE('2026-03-10 15:00','YYYY-MM-DD HH24:MI'),
TO_DATE('2026-03-10 16:52','YYYY-MM-DD HH24:MI'))
  4    INTO Jadwal (id_jadwal, id_film, id_theater, periode_start, periode_end)
VALUES (503, 3, 103, TO_DATE('2026-03-10 17:00','YYYY-MM-DD HH24:MI'),
TO_DATE('2026-03-10 19:02','YYYY-MM-DD HH24:MI'))
  5    INTO Jadwal (id_jadwal, id_film, id_theater, periode_start, periode_end)
VALUES (504, 4, 104, TO_DATE('2026-03-11 19:00','YYYY-MM-DD HH24:MI'),
TO_DATE('2026-03-11 20:03','YYYY-MM-DD HH24:MI'))
  6    INTO Jadwal (id_jadwal, id_film, id_theater, periode_start, periode_end)
VALUES (505, 5, 105, TO_DATE('2026-03-11 21:00','YYYY-MM-DD HH24:MI'),
TO_DATE('2026-03-11 21:46','YYYY-MM-DD HH24:MI'))
  7  SELECT * FROM dual;
SQL> INSERT ALL
  2    INTO Transaksi (kode_pemesanan, id_jadwal, id_member, status, tanggal,
total_harga) VALUES ('TX01', 501, 1, 'LUNAS', TO_DATE('2026-03-08','YYYY-
MM-DD'), 45000)
  3    INTO Transaksi (kode_pemesanan, id_jadwal, id_member, status, tanggal,
total_harga) VALUES ('TX02', 502, 2, 'LUNAS', TO_DATE('2026-03-08','YYYY-
MM-DD'), 50000)
  4    INTO Transaksi (kode_pemesanan, id_jadwal, id_member, status, tanggal,
total_harga) VALUES ('TX03', 503, 3, 'LUNAS', TO_DATE('2026-03-08','YYYY-
MM-DD'), 75000)
  5    INTO Transaksi (kode_pemesanan, id_jadwal, id_member, status, tanggal,
total_harga) VALUES ('TX04', 504, 4, 'PROSES', TO_DATE('2026-03-08','YYYY-
MM-DD'), 100000)
  6    INTO Transaksi (kode_pemesanan, id_jadwal, id_member, status, tanggal,
total_harga) VALUES ('TX05', 505, 5, 'LUNAS', TO_DATE('2026-03-08','YYYY-
MM-DD'), 150000)
  7  SELECT * FROM dual;
```

Penjelasan :

Query diatas menggunakan perintah INSERT ALL agar pengisian data lebih efisien (sekaligus banyak). Syarat utamanya adalah data pada tabel Induk harus diisi lebih dulu sebelum data pada tabel Anak.

3. Lakukan update data (minimal 5 data), tuliskan script SQL nya dan sertakan tangkapan layarnya.

   Screenshot :

```
SQL> UPDATE Film
  2  SET sinopsis = 'Kisah legendaris pertukaran tubuh Taki dan Mitsuha'
  3  WHERE id_film = 1;
SQL> UPDATE Theater
  2  SET harga = 65000
  3  WHERE id_theater = 103;
SQL> UPDATE Member
  2  SET email = 'suzume.iwato@newmail.jp', no_hp = '08771122'
  3  WHERE id_member = 4;
SQL> UPDATE Jadwal
  2  SET periode_start = TO_DATE('2026-03-15 20:00', 'YYYY-MM-DD HH24:MI')
  3  WHERE id_jadwal = 504;
SQL> UPDATE Transaksi
  2  SET status = 'LUNAS'
  3  WHERE kode_pemesanan = 'TX04';
SQL> SELECT judul, sinopsis FROM Film WHERE id_film = 1;

JUDUL                    SINOPSIS
------------------------ ------------------------------
Your Name                Kisah legendaris pertukaran tu
                         buh Taki dan Mitsuha

SQL> SELECT kelas, harga FROM Theater;

KELAS               HARGA
--------------- ----------
Reguler              45000
Reguler Plus         50000
Dolby                65000
Premiere            100000
Velvet              150000
```

   Jawaban :

   SQL> UPDATE Film
   2 SET sinopsis = 'Kisah legendaris pertukaran tubuh Taki dan Mitsuha'
   3 WHERE id_film = 1;
   SQL> UPDATE Theater
   2 SET harga = 65000
   3 WHERE id_theater = 103;
   SQL> UPDATE Member
   2 SET email = 'suzume.iwato@newmail.jp', no_hp = '08771122'
   3 WHERE id_member = 4;
   SQL> UPDATE Jadwal
   2 SET periode_start = TO_DATE('2026-03-15 20:00', 'YYYY-MM-DD HH24:MI')
   3 WHERE id_jadwal = 504;
   SQL> UPDATE Transaksi
   2 SET status = 'LUNAS'
   3 WHERE kode_pemesanan = 'TX04';

   Penjelasan :

Perintah UPDATE selalu membutuhkan klausa WHERE agar perubahan tidak mengenai seluruh isi tabel. Contohnya, mengubah harga tiket pada kelas tertentu saja atau memperbarui email member.

4. Lakukan hapus data (minimal 2 data), tuliskan script SQL nya dan sertakan tangkapan layarnya.

   Screenshot :

```
SQL> DELETE FROM Transaksi
  2  WHERE id_member = (SELECT id_member FROM Member WHERE no_hp = '0812')
  3  ;
SQL> DELETE FROM Member
  2  WHERE no_hp = '0812';
SQL> select * from member;

 ID_MEMBER NAMA            NO_HP                TGL_LAHIR EMAIL
---------- --------------- -------------------- --------- --------------------
         1 Taki            0811                 01-DEC-99 taki@mail.com
         3 Hodaka          0813                 10-JAN-05 hodaka@mail.com
         4 Suzume          08771122             24-MAY-06 suzume.iwato@newmail
                                                          .jp

         5 Takaki          0815                 15-MAR-85 takaki@mail.com
```

   Jawaban :
   SQL> DELETE FROM Transaksi
   2 WHERE id_member = (SELECT id_member FROM Member WHERE no_hp = '0812')
   3 ;
   SQL> DELETE FROM Member
   2 WHERE no_hp = '0812';

   Penjelasan :
   SQL melarang penghapusan data Induk (Member) jika masih ada data Anak (Transaksi) yang merujuk padanya. Jadi, prosedurnya adalah hapus dulu transaksinya, baru hapus membernya.